

Enhancing the Accuracy of Heart Disease Prediction: A Comparative Study of Random Forest and Gradient Boosting Models Based on Physiological Indicators

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Abstract—Heart disease is a prevalent and critical health issue, making its prediction a major focus in medical research. However, traditional predictive models often face challenges such as low accuracy, poor model fitting, and limited robustness. To address these issues, this study improves upon traditional decision tree-based models by introducing two ensemble learning approaches: Random Forest and Gradient Boosting Tree. These methods significantly enhance prediction accuracy and effectively reduce the risk of overfitting. Using a dataset of over 1,300 patient records from the Kaggle platform, a series of data preprocessing techniques were applied to improve data stability. Experimental results demonstrate that the proposed methods achieve high accuracy and precision in heart disease risk prediction. This research provides a robust analytical tool for clinical decision support and early screening of heart disease, offering valuable potential for practical application.

Keywords—ensemble learning models, feature extraction, interactive dashboard, feature visualization

I. INTRODUCTION

In recent years, the global incidence of heart disease has continued to rise, with an increasingly younger trend, making heart disease prediction a persistent focus for researchers. Although some progress has been made [2][3][4], the accuracy of current prediction results remains unsatisfactory. Inaccurate predictions of heart disease can lead to many serious and irreversible consequences.

However, most existing studies rely on traditional prediction models, such as single decision trees[1] or linear regression. These models often suffer from limited prediction accuracy, poor stability, and weak generalization ability, making them prone to overfitting and unable to effectively capture the complex nonlinear relationships in physiological data. Furthermore, many traditional approaches require extensive domain knowledge for feature engineering, which restricts their adaptability and scalability in practical applications.

With the rapid development of machine learning, ensemble learning models such as Random Forest[5][8] and Gradient Boosting Tree[6] have emerged, combining the predictions of multiple base learners to improve accuracy and robustness. Despite their advantages, current research still faces challenges in model selection, comprehensive evaluation, and the interpretability of results. In particular, there is a lack of systematic comparison between different ensemble models [7] on heart disease datasets, and few studies provide interactive tools for in-depth data exploration and model analysis.

In this study, we developed a heart disease prediction system that employs two models for simultaneous prediction, using patient feature data to determine the likelihood of heart disease. Our main contributions are as follows:

We constructed a heart disease prediction framework based on Random Forest and Gradient Boosting Tree models, utilizing physiological indicators for analysis. Additionally, we developed an interactive dashboard using Streamlit [9], supporting comprehensive data exploration, model evaluation, and result visualization. Through specific hyperparameter configuration, cross-validation, multi-metric comparison, and consistency analysis, we thoroughly evaluated and compared model performance. By integrating data preprocessing (IQR-based outlier handling and missing value processing) and feature engineering (such as model-based feature selection for Random Forest), we enhanced the effectiveness and reliability of the predictions.

The remainder of this paper is organized as follows: Section 2 reviews related work in heart disease prediction. Section 3 describes the dataset and data preprocessing methods. Section 4 details the construction and rationale of the Random Forest and Gradient Boosting Tree models. Section 5 presents the experimental setup and results, including feature visualization and model performance evaluation. Section 6 concludes the paper and discusses future research directions.

II. RELATED WORK

Traditional methods for heart disease risk prediction [10] mainly rely on statistical models or single machine learning algorithms (such as logistic regression and decision trees). However, these approaches have limited ability to handle nonlinear interactions in physiological data and require extensive domain knowledge for feature engineering [11], which restricts prediction accuracy and generalization capability.

Table 1 Heart Disease Prediction Dataset

Number	Age	Gender	Heart rate	Systolic blood pressure	Diastolic blood pressure	Blood sugar	CK-MB	Troponin	Result
1	64	male	66	160	83	160	1.8	0.012	negative
2	21	male	94	98	46	296	6.75	1.06	positive
3	55	male	64	160	77	270	1.99	0.003	negative
4	64	male	70	120	55	270	13.87	0.122	positive

In recent years, ensemble learning methods—especially tree-based models such as Random Forest (RF) and Gradient Boosting Machine (GBM)—have achieved significant progress in the biomedical field.

Random Forest reduces model variance and enhances robustness by constructing multiple independent decision trees and aggregating their predictions. Gradient Boosting Tree, on the other hand, iteratively builds weak learners to progressively correct the residuals of previous models, thereby reducing bias and excelling at handling complex patterns and feature interactions. These ensemble models perform exceptionally well in capturing nonlinear relationships and interactions between features, making them superior for tasks such as heart disease prediction.

Despite the powerful capabilities of ensemble models, their performance is highly dependent on the quality of input data and comprehensive model evaluation. In medical data analysis, data preprocessing (such as handling missing values, correcting outliers, and feature selection) is crucial. In addition, model evaluation should comprehensively consider multiple metrics [17] (such as precision, recall, F1-score, and AUC) and incorporate cross-validation to ensure robustness. Resnet has also been evaluated on Detection of Arrhythmia [18].

III. DATASET

A. Dataset Overview

The dataset used in this study, Rashid, Tarik A.; Hassan, Bryar (2022), “Heart Attack Dataset”, Mendeley Data, V1, doi:10.17632/wmhctcrt5v.1, was obtained from the open-source Kaggle platform. This dataset includes the following variables: gender, age, heart rate, systolic blood pressure, diastolic blood pressure, blood sugar, creatine kinase-MB,

troponin, and outcome. Each patient’s record consists of these nine categories. The entire dataset contains 1,320 samples, originally collected at Zheen Hospital in Erbil, Iraq, between January and May 2019. Table 1 below provides a selection of feature data from this dataset.

B. Data Preprocessing

For handling missing values, this study defines a function named `dropna()`, which removes all rows in the dataset containing any missing values (NaN). This is a straightforward approach that ensures the model will not encounter issues caused by missing values during training.

For outlier processing, a set of specified numerical features (numeric_cols = ['Age', 'HeartRate', 'SysBloodPres', 'DiaBloodPres', 'BloodSugar', 'CK-MB', 'Troponin']) is selected, and the code applies the Interquartile Range (IQR) [12][13] method for identification and handling.

The interquartile range is calculated ($IQR = Q3 - Q1$) (1) as shown in formula (1). The lower bound for outliers is defined as ($lower = Q1 - 1.5 * IQR$), formula (2), and the upper bound as ($upper = Q3 + 1.5 * IQR$), formula (3). Any value below the lower bound is replaced with the lower bound, and any value above the upper bound is replaced with the upper bound. This method, known as “winsorization” or boundary replacement, aims to reduce the potential negative impact of extreme values on model training.

IV. MODELS

A. Random Forest and Gradient Boosting Tree

The Random Forest model is first used for feature selection [14]. Specifically, `SelectFromModel` is combined with the median threshold of feature importance to select the optimal subset of features from the training data. The main Random Forest classifier is then trained on these selected features with empirically set parameters (`RandomForestClassifier(n_estimators=200, max_depth=10, min_samples_split=5)`), which effectively reduces model training time.

The Gradient Boosting Tree model is trained on the complete feature set, also using empirically set parameters (`GradientBoostingClassifier(n_estimators=150, learning_rate=0.2, max_depth=4, subsample=0.7)`).

Both models are evaluated using accuracy, precision, recall, F1-score, and AUC (area under the ROC curve) on their respective test sets, as well as 5-fold cross-validation accuracy on the entire dataset. Feature importance for each model is also extracted for further analysis. All results and metrics are presented to users via an interactive dashboard built with Streamlit.

To enhance the scientific rigor of the methodology, the following mathematical formulations are included:

Random Forest Prediction:

$$\hat{y} = \text{majority}_{\text{vote}}(h_1(x), h_2(x), \dots, h_T(x)) \quad (1)$$

$h_T(x)$: The prediction result of the t -th tree for the input sample x . Each tree is an independent classifier trained on

different subsets of data and features.

$\hat{y}$: The final predicted class for the input sample.

T: The number of decision trees in the random forest.

Gradient Boosting Tree Prediction:

$$\hat{y} = \sum_{m=1}^M \gamma_m h_m(x) \quad (2)$$

$\hat{y}$: The final predicted value for the input sample x .

M: The total number of decision trees constructed during the boosting process.

γ_m : The learning rate, which controls the influence of each tree on the final prediction. The learning rate is usually a small positive number used to adjust the contribution of each newly added tree to the overall model.

$h_m(x)$: The output of the m -th tree for the input sample x . Each tree attempts to correct the errors of the previous tree, gradually reducing the overall error of the model.

B. Model Selection and Performance Comparison

The two machine learning models may perform differently on various heart disease datasets. This approach allows for a direct comparison of their performance (e.g., accuracy, precision, recall, F1-score, AUC) under different datasets, enabling targeted decisions on which specific model to use and providing more options for healthcare professionals. For example, in this dataset, the Gradient Boosting Tree outperforms Random Forest in all aspects, but in datasets with more noise, smaller sample sizes, or altered feature relationships, Random Forest may achieve higher prediction accuracy.

C. Enhancing the Confidence of Results

The two models differ significantly in structure and learning mechanism. Random Forest constructs multiple independent decision trees in parallel, each trained on a randomly sampled data subset, and combines predictions by majority vote to reduce variance. In contrast, Gradient Boosting Tree builds trees sequentially, with each new tree correcting the errors of the previous ensemble, focusing on reducing bias. If both models yield similar predictions on a common test set, it increases confidence in the results and demonstrates robustness.

D. Training and Evaluation

Data preprocessing is performed, including handling missing values and outliers. The data is then split into training and test sets for both models using different random seeds. Random Forest additionally conducts feature selection, while Gradient Boosting Tree uses all features. Both models are trained independently and evaluated using 5-fold cross-validation accuracy, test set accuracy, precision, recall, F1-score, and AUC. All results, evaluation metrics, feature importance, and model comparison analyses are visualized in the Streamlit dashboard.

V. EXPERIMENTS

A. Experimental Setup

All experiments were conducted in a Python 3.12.9

environment using an NVIDIA GeForce RTX 4060 GPU. The core dependencies include Scikit-learn (for model training, feature selection, cross-validation, and performance metrics), Pandas (for data processing and analysis), NumPy (for numerical computation), Matplotlib and Seaborn (for data visualization), and Streamlit (for building interactive web application dashboards). More detailed model performance was comprehensively evaluated using accuracy, confusion matrix, classification report (including precision, recall, and F1-score), and area under the ROC curve (AUC).

B. Feature Visualization

The result demonstrates comprehensive and diverse capabilities in feature visualization [15][16], providing users with tools to gain in-depth understanding of data features and their relationship with heart disease risk through multiple tabs in the Streamlit dashboard. Specifically:

1) Feature Distribution Visualization

In the “Data Overview” tab, a histogram is generated for each feature in the dataset (including numerical features and

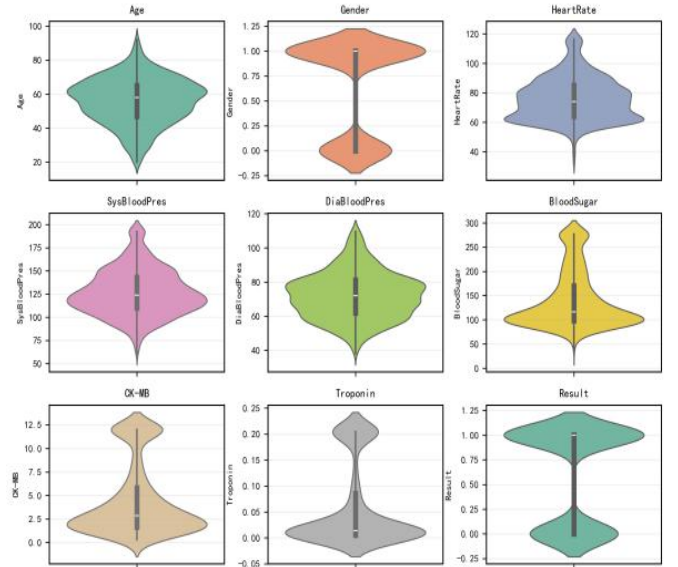


Fig. 2. Histogram of data feature distributions with overlaid kernel density estimation (KDE) curves.

the transformed target variable), overlaid with a kernel density estimation (KDE) curve, as shown in Fig. 2, including the value distribution of each physiological indicator, such as whether it is skewed, normally distributed, or multimodal, as well as the general range and central tendency of the data.

2) Visualization of the Relationship Between Features and the Target Variable

In the “Feature Analysis” tab, users can select any feature and generate a box plot of that feature across different heart disease risk categories (positive/negative), as shown in Fig. 3, which clearly displays the distribution differences of the selected physiological indicator among different risk groups, such as median, interquartile range, and outliers, which helps to assess the potential of different features to distinguish heart disease risk.

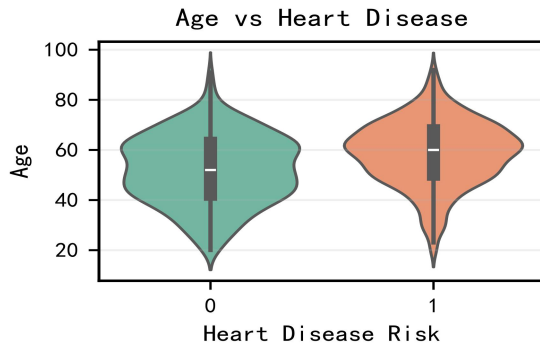


Fig. 3. Box plot illustrating the relationship between features and the target variable.

3) Visualization of Relationships Between Features

In the “Feature Analysis” tab, users can select any two features to plot a scatter plot, with different colors indicating heart disease risk categories, as shown in Fig. 4, where shows the joint distribution of two physiological indicators and assess whether this joint distribution can better separate different risk groups, thereby uncovering potential feature interactions.

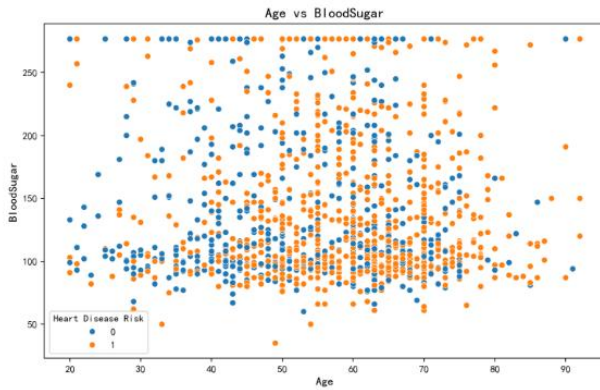


Fig. 4. Scatter plot illustrating the relationships between features.

4) Feature Correlation Visualization

In the “Correlation Analysis” tab, the Pearson correlation coefficient matrix among all numerical features (including the target variable) is calculated and displayed as a heatmap. Additionally, the features most strongly correlated with the target variable and pairs of features with high inter-correlation are specifically listed, as shown in Fig. 5, which intuitively reveals the strength and direction of linear relationships among different physiological indicators, as well as which indicators have the closest linear relationship with heart disease risk. It helps in understanding issues of multicollinearity and potential feature redundancy.

5) Feature Importance Visualization

Capability: In the “Feature Importance” tab, the importance of each feature as calculated by the Random Forest and Gradient Boosting Tree models is displayed in both horizontal bar charts and tables, as shown in Fig. 6, which clearly identifies which physiological indicators contribute most to heart disease risk prediction for the two different ensemble learning models, providing important references for feature selection and understanding the basis of model decisions.

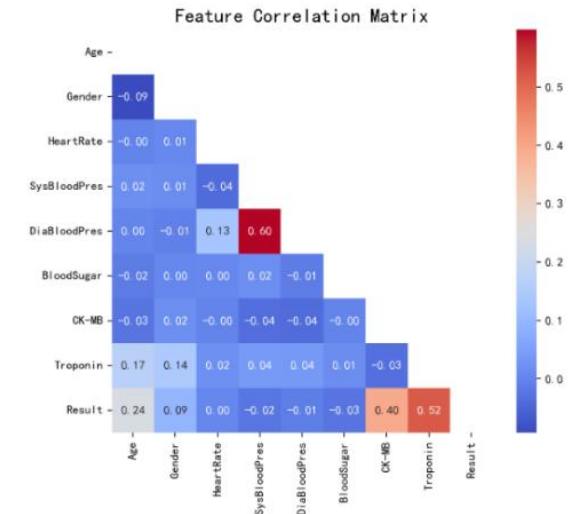


Fig. 5. Pearson correlation coefficient matrix among feature data

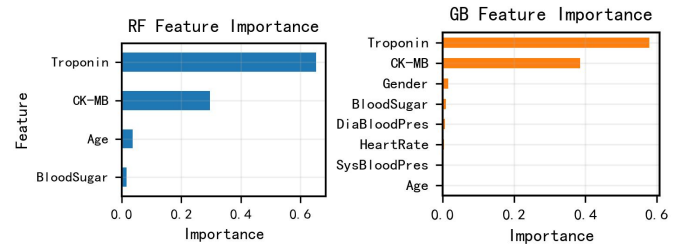


Fig. 6. Bar chart of feature importance

Through these diverse visualization methods, this study not only presents the characteristics of the raw data, but also deeply analyzes the relationships between features and the target variable, between features themselves, and the feature importance from the model perspective, providing users with a powerful platform to understand the data and model behavior from multiple angles.

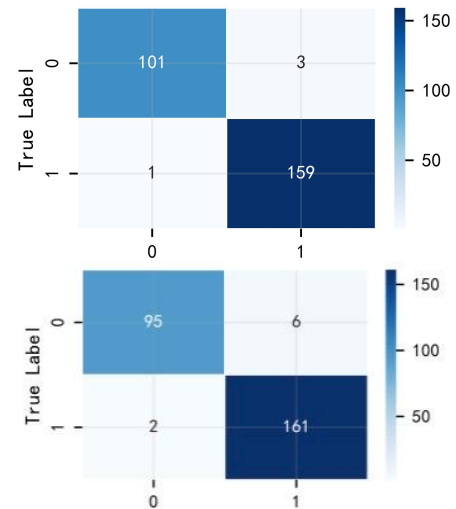


Fig. 7. Confusion matrices of Random Forest and Gradient Boosting Tree

C. Classification Results and Confusion Matrix

The “Model Performance” tab of the Streamlit dashboard displays the confusion matrices of the Random Forest and

Gradient Boosting models on the test set, the results in Fig. 7, intuitively presenting the distribution of model predictions for heart disease risk (positive/negative). Users can evaluate the model's performance on both classes; the diagonal elements of the confusion matrix represent the number of correctly classified samples, while off-diagonal elements indicate misclassifications.

The classification report table summarizes the precision, recall, and F1-score for each class, helping healthcare professionals assess the model's ability to identify specific categories. If the F1-score for certain classes is low, possible reasons include high feature similarity or insufficient sample size. The "Feature Analysis," "Correlation Analysis," and "Feature Importance" tabs of the dashboard help healthcare professionals explore the reasons for performance differences.

Overall accuracy and other key metrics are clearly displayed in the tabs. By observing the confusion matrix, healthcare professionals can determine whether the model makes correct classifications in most cases. When the model is confused, the off-diagonal elements become more prominent, indicating a higher number of misclassified samples. These classification results and confusion matrices help healthcare professionals gain deeper insights into the strengths and weaknesses of the model, providing directions for improvement, such as adjusting feature selection strategies or optimizing model parameters.

VI. CONCLUSION

This study proposes a series of effective methods in the field of heart disease prediction. Through the training and evaluation of Random Forest and Gradient Boosting Tree models, high accuracy and precision in model prediction were achieved. Experimental results show that the Random Forest model reached an accuracy of 96.97% and a precision of 96.41%, while the Gradient Boosting Tree achieved an accuracy of 98.48% and a precision of 98.15%.

Furthermore, as shown by the ROC curves and AUC in Figure 1, the performance of the Gradient Boosting Tree is slightly superior to that of the Random Forest model on this dataset. Regarding the correlation between feature data and prediction results: the Random Forest model identified troponin as having a correlation of 65.26%, creatine kinase-MB 29.58%, age 3.61%, and blood sugar 1.56%. The Gradient Boosting Tree model indicated troponin with a correlation of 57.92%, creatine kinase-MB 38.54%, gender 1.56%, and blood sugar, diastolic blood pressure, heart rate, systolic blood pressure, and age all below 1%.

In summary, the results demonstrate the effectiveness of applying and systematically comparing different traditional machine learning strategies, the Random Forest method with feature selection and the Gradient Boosting method using the full feature set, for heart disease risk prediction tasks, and provide a practical solution for comprehensive data exploration, model performance evaluation (including confusion matrix, classification report, and ROC curve comparison), and in-depth understanding of model prediction behavior (such as prediction consistency analysis and feature importance comparison),

offering a valuable analytical tool for decision support in clinical settings.

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